

FLUID MECHANICS (ME-205)



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Unit-I I

Fluid Dynamics and Dimensional Analysis

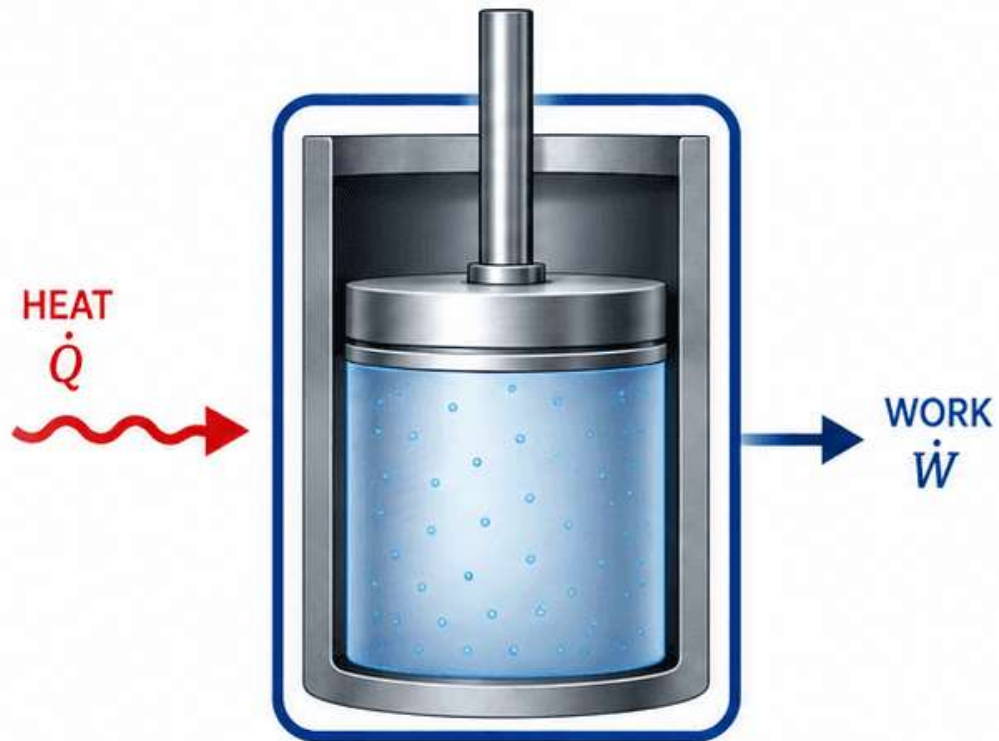
Topic Covered

- Concept of system and control volume;
- Reynolds's transport theorem,
- Euler's equation,
- Bernoulli's equation,
- Navier-Stokes equation;
- Flow measurement: Venturi meter, Orifice meter, Pitot-tube, flow meters, notches.
- Buckingham's π Theorem.
- Non-dimensional parameters,
- similarity and its application to fluid problems.

Concept of system and control volume

CONTROL MASS

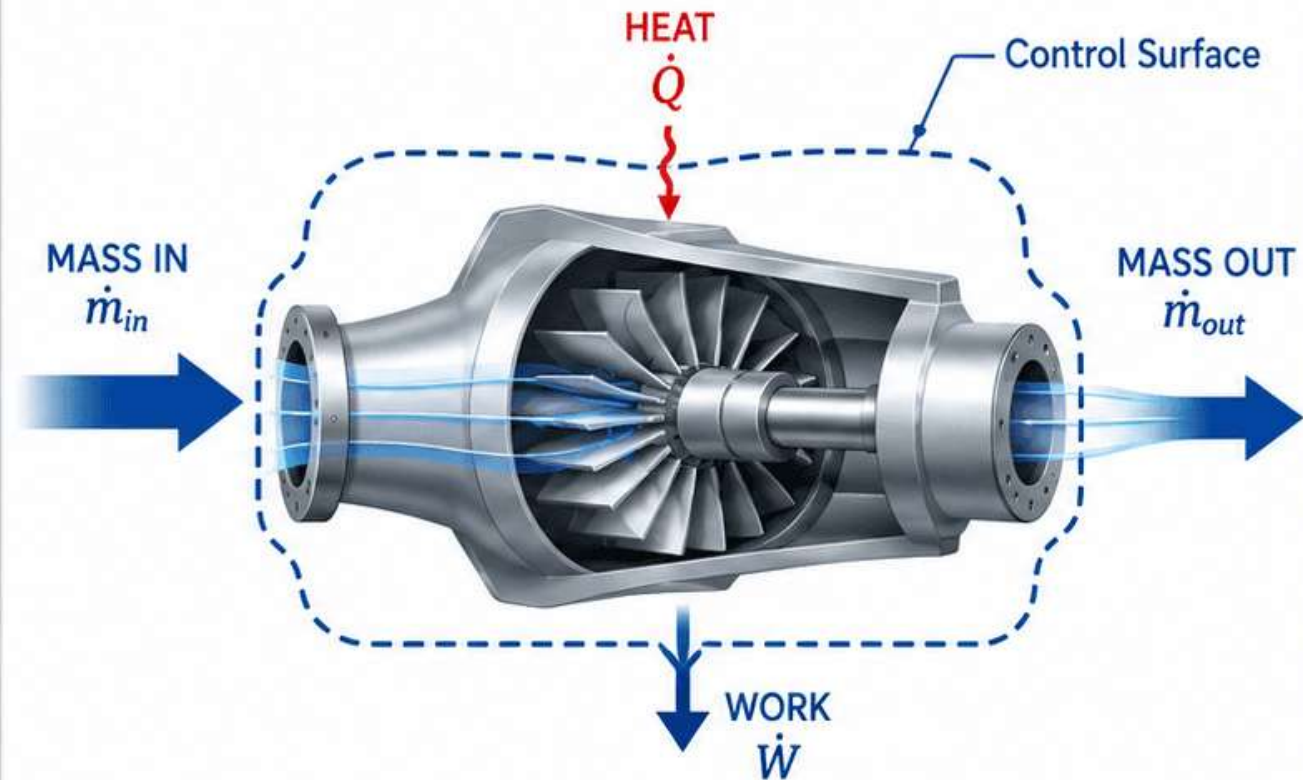
Closed System



Fixed mass
No mass crosses boundary.

CONTROL VOLUME

Open System



Mass may enter and leave.

Concept of system and control volume

System: - an arbitrary quantity of mass of fixed identity.

Surrounding: - Anything external to the system.

Mass conservation

$$m_{sys} = \text{const}$$
$$\frac{dm}{dt} = 0$$

Linear momentum Equation

$$F = ma = \frac{mdV}{dt} = \frac{d(mV)}{dt}$$

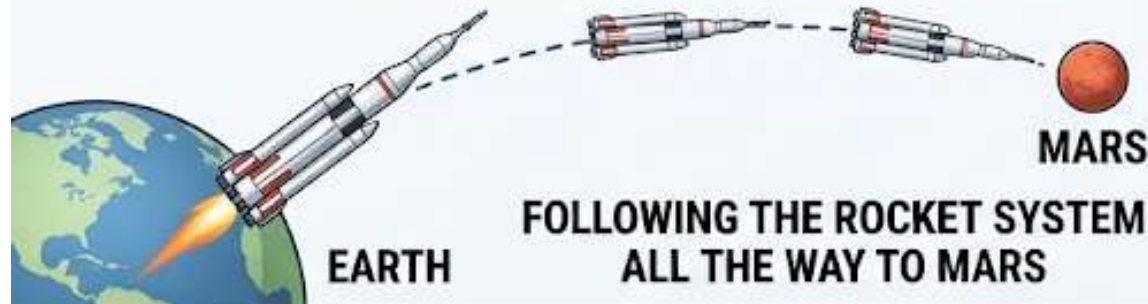
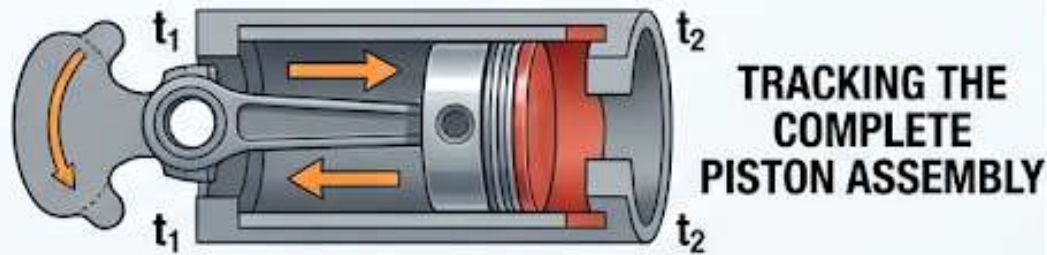
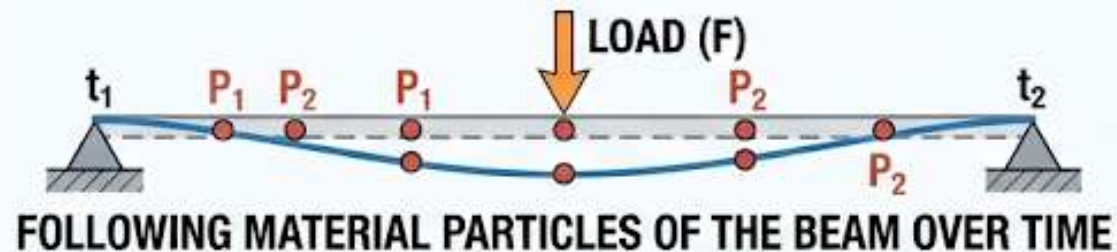
Energy Equation

$$\frac{dE}{dt} = \dot{Q} - \dot{W}$$

Concept of system and control volume

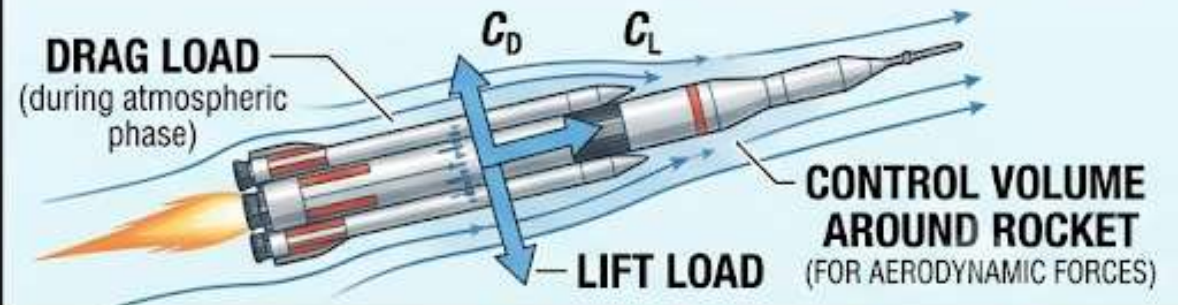
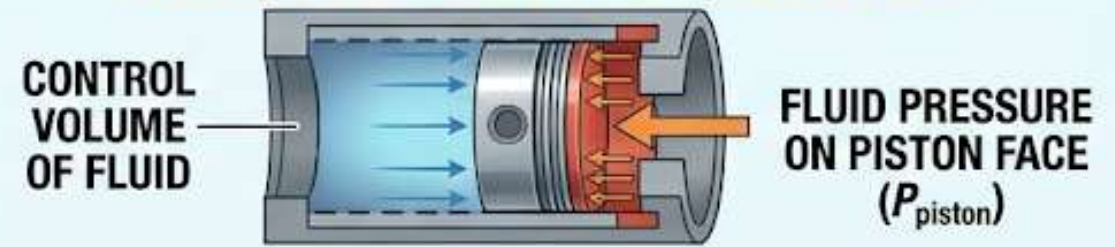
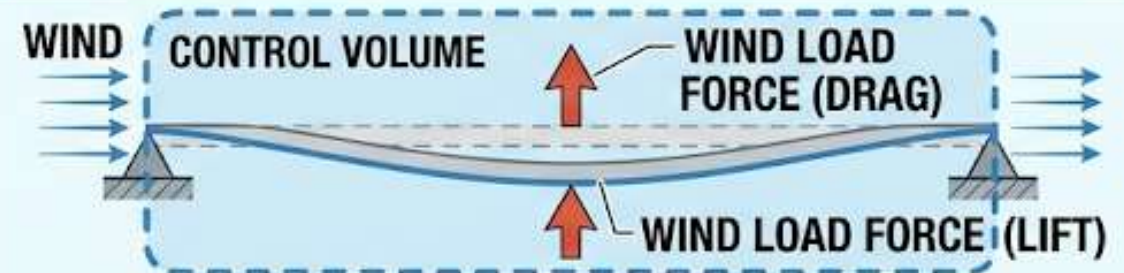
COMPARING APPROACHES: SYSTEM vs. CONTROL VOLUME ANALYSIS

SYSTEM APPROACH (LAGRANGIAN): FOLLOWING SPECIFIC OBJECTS



FOCUS ON A **CLOSED, FIXED IDENTITY SYSTEM**.
We track specific particles or objects over time.

CONTROL VOLUME APPROACH (EULERIAN): LOCALIZED FLUID INTERACTIONS



FOCUS ON AN **OPEN, FIXED REGION IN SPACE**.
We analyze interactions of fluids passing through a volume.

Concept of system and control volume

For example, in the case of the system, we follow

- A beam as it deflects under load.*
- A piston as it oscillates.*
- A rocket system all the way to Mars.*

In fluid systems, for the three examples just cited, we wish to know

- the wind loads on the beam,*
- the fluid pressures on the piston,*
- the drag and lift loads on the rocket.*

But fluid systems do not demand this concentrated attention.

We are not interested in the ultimate path of a specific fluid particle.

Instead, we wish to understand how fluid flow influences the product we are interested in.

In analysing a control volume, we apply the system laws to a specific region that the system may occupy only for an instant. The system passes on, and other systems come along.

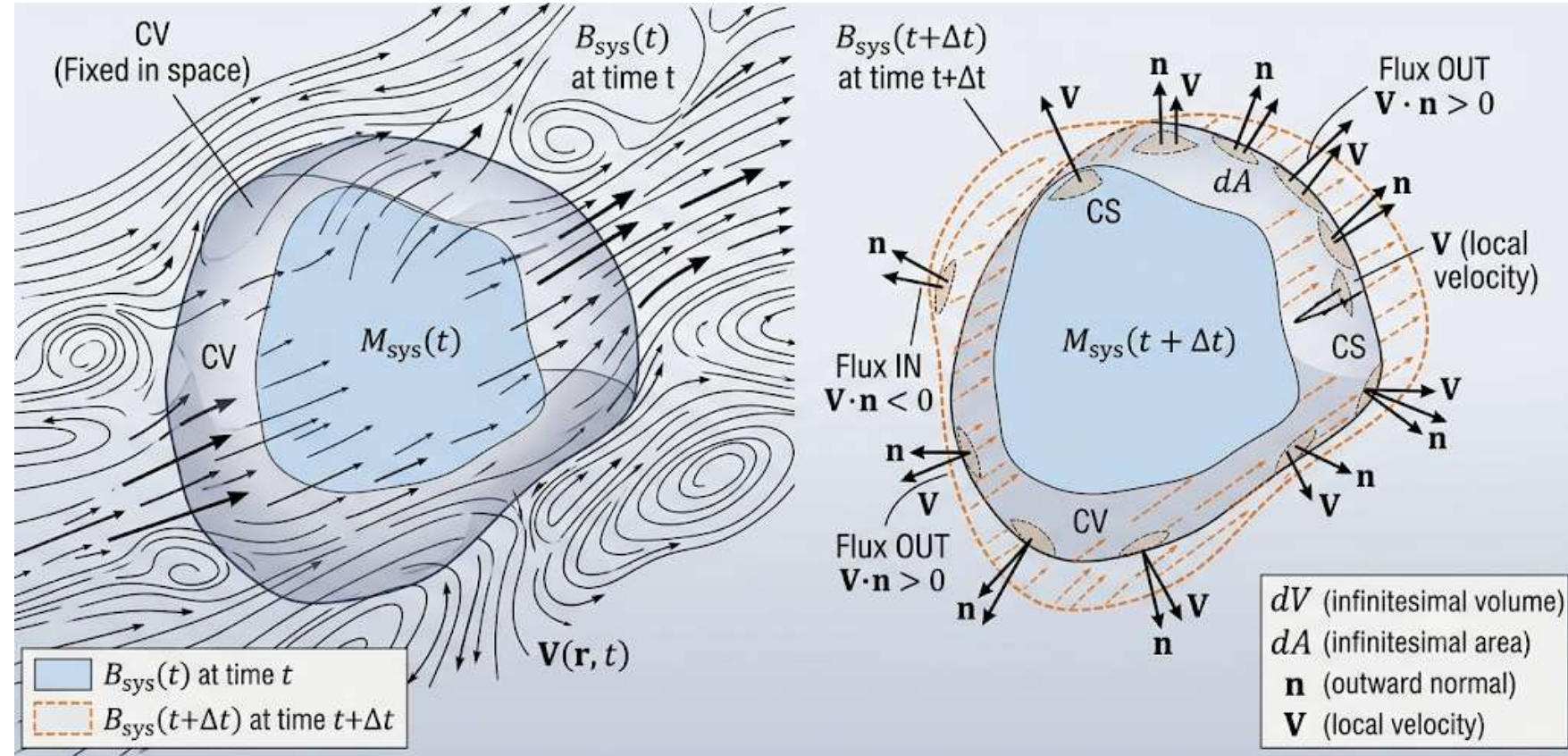
Reynold Transport theorem

To convert a system analysis to a control volume analysis, we must build a bridge that connects the mathematics of a system of individual masses to a specific region defined as the control volume. This bridge or conversion, called the **Reynolds transport theorem**, can be applied to all the basic laws.

We need to relate the time derivative of a system property to the rate of change of that property within a control volume

Reynold Transport theorem

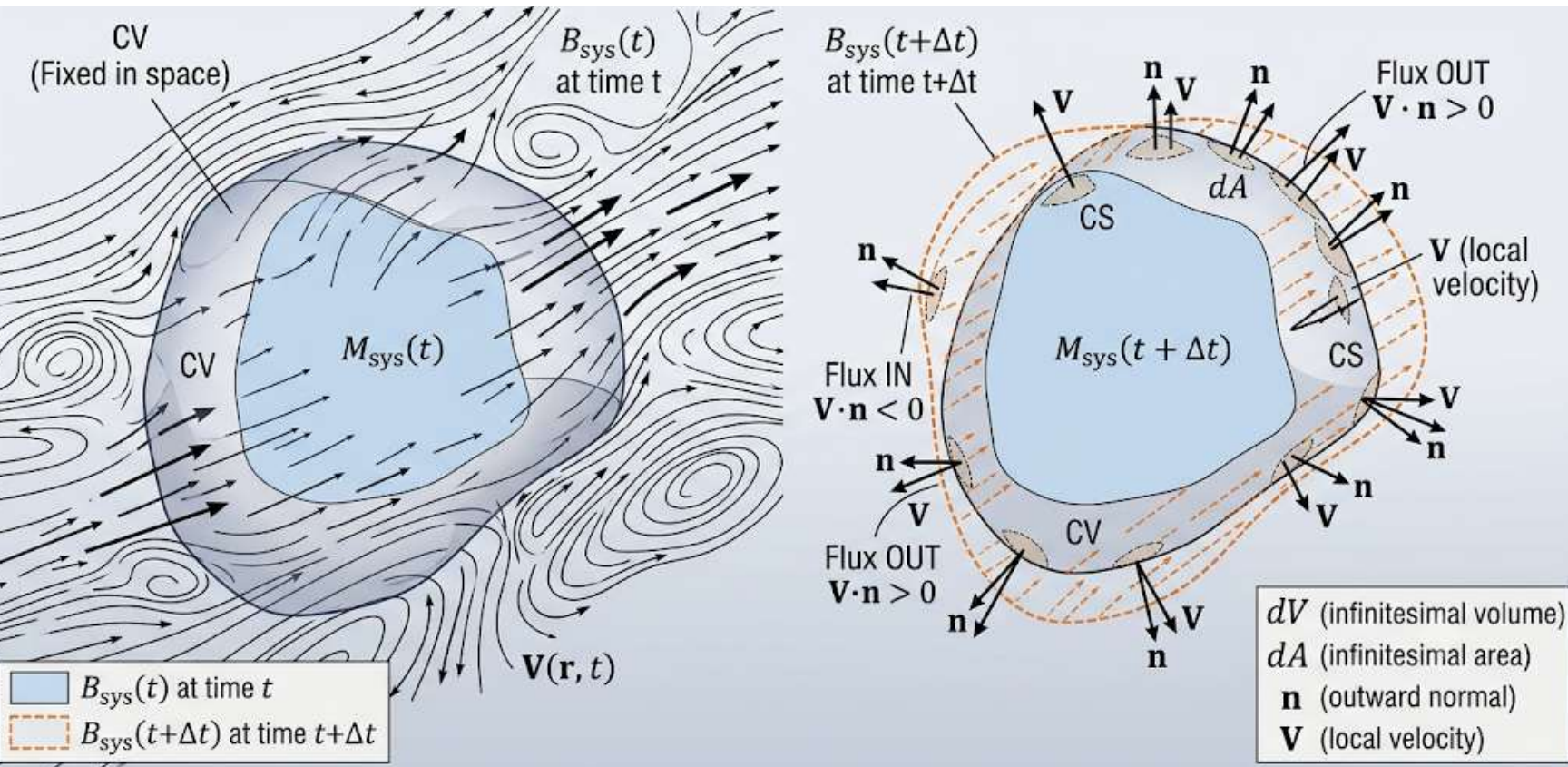
Let B be any property of the fluid (energy, momentum, enthalpy, etc.) and let $\beta = \frac{dB}{dm}$ be the intensive value, or the amount of B per unit mass in any small element of the fluid. The total amount of B in the control volume (the solid curve) is thus



$$B_{CV} = \int \beta dm = \int \beta \rho dV ; \beta = \frac{dB}{dm}$$

Reynold Transport theorem

changes in B relating to the control volume:



Change within CV

$$\frac{d}{dt} \left(\int_{CV} \beta \rho dV \right)$$

Outflow of β from CV

$$\left(\int_{CS} \beta \rho V \cos \theta dA_{out} \right)$$

Inflow of β to CV

$$\left(\int_{CS} \beta \rho V \cos \theta dA_{in} \right)$$

$$\frac{d}{dt} (B_{sys}) = \frac{d}{dt} \left(\int_{CV} \beta \rho dV \right) + \left(\int_{CS} \beta \rho V \cos \theta dA_{out} \right) - \left(\int_{CS} \beta \rho V \cos \theta dA_{in} \right)$$

Reynold Transport theorem

$$\frac{d}{dt}(B_{sys}) = \frac{d}{dt}\left(\int_{CV} \beta \rho d\mathcal{V}\right) + \left(\int_{CS} \beta \rho V \cos\theta dA_{out}\right) - \left(\int_{CS} \beta \rho V \cos\theta dA_{in}\right)$$

$$\frac{d}{dt}(B_{sys}) = \frac{d}{dt}\left(\int_{CV} \beta \rho d\mathcal{V}\right) + \left(\int_{CS} \beta \rho (V \cdot n) dA\right)$$

where, Flow terms = $\left(\int_{CS} \beta \rho (V \cdot n) dA\right)$

If the control volume is moving uniformly at velocity V_s

$$V_r = V - V_s$$

$$\frac{d}{dt}(B_{sys}) = \frac{d}{dt}\left(\int_{CV} \beta \rho d\mathcal{V}\right) + \left(\int_{CS} \beta \rho (V_r \cdot n) dA\right)$$

Reynold Transport theorem

$$\frac{d}{dt}(B_{sys}) = \frac{d}{dt}\left(\int_{CV} \beta \rho d\mathcal{V}\right) + \left(\int_{CS} \beta \rho (V_r \cdot n) dA\right)$$

For fixed control volume and $V_r = V$

For mass conservation $B = m, \beta = 1$

$$\left(\int_{CV} \frac{\partial \rho}{\partial t} d\mathcal{V}\right) + \left(\int_{CS} \rho (V \cdot n) dA\right) = 0$$

For linear momentum $B = mV, \beta = V$

$$\frac{d}{dt}(mV)_{sys} = \Sigma F = \frac{d}{dt}\left(\int_{CV} V \rho d\mathcal{V}\right) + \left(\int_{CS} V \rho (V \cdot n) dA\right)$$

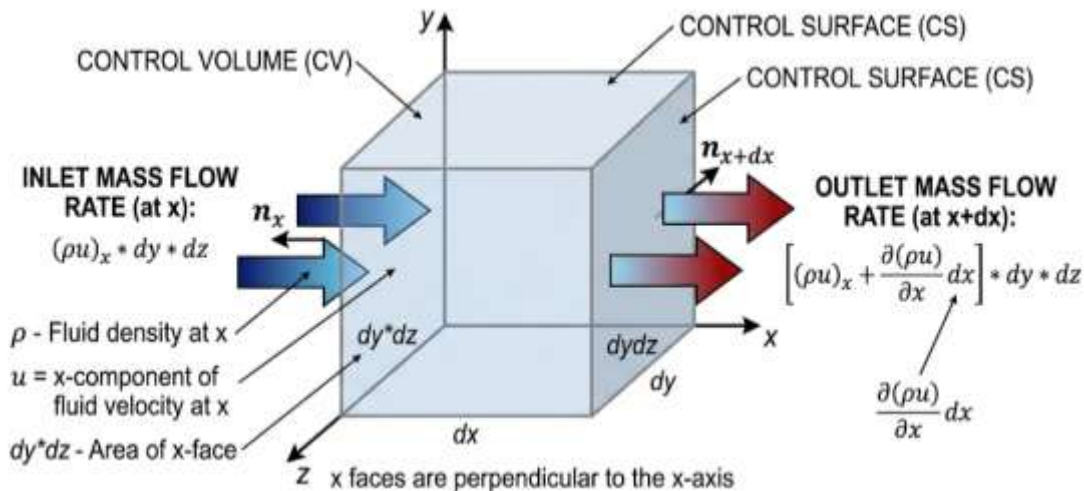
Euler Equation

Assumptions: -

- Flow is steady and inviscid.
- Flow is along a streamline.
- Fluid is incompressible (density ρ is constant).

$$\Sigma F = \frac{\partial}{\partial t} \left(\int_{CV} V \rho d\mathcal{V} \right) + \left(\int_{CS} V \rho (V_r \cdot n) dA \right)$$

Euler Equation



$$\Sigma F_{CV} = \frac{\partial}{\partial t} \left(\int_{CV} V \rho dV \right) + \left(\int_{CS} V \rho (V_r \cdot n) dA \right)$$

Define the Forces

$$\Sigma F_{CV} = \left(\int_{CV} \rho g dV \right) - \left(\int_{CS} P n dA \right)$$

Using the Gradient Theorem, we convert the surface integral into a volume integral:

$$- \int_{CS} P n dA = - \left(\int_{CV} \nabla P dV \right) \quad \text{and} \quad \int_{CS} V \rho (V_r \cdot n) dA = \int_{CV} \nabla \cdot (\rho V V) dV$$

$$\left(\int_{CV} \rho g dV \right) - \left(\int_{CV} \nabla P dV \right) = \left(\int_{CV} \frac{\partial}{\partial t} (\rho V) dV \right) + \int_{CV} \nabla \cdot (\rho V V) dV$$

$$\left(\int_{CV} (\rho g - \nabla P) dV \right) = \left(\int_{CV} \frac{\partial}{\partial t} (\rho V) dV \right) + \int_{CV} \nabla \cdot (\rho V V) dV$$

Euler Equation

$$\left(\int_{CV} (\rho g - \nabla P) d\mathcal{V} \right) = \left(\int_{CV} \frac{\partial}{\partial t} (\rho V) d\mathcal{V} \right) + \int_{CV} \nabla \cdot (\rho V V) d\mathcal{V}$$

$$\int_{CV} \left[\frac{\partial}{\partial t} (\rho V) + \nabla \cdot (\rho V V) + \nabla P - \rho g \right] d\mathcal{V} = 0$$

Since this must hold true for *any* arbitrarily chosen control volume, the integrand itself must be zero:

$$\frac{\partial}{\partial t} (\rho V) + \nabla \cdot (\rho V V) = -\nabla P + \rho g$$

Simplifying using the continuity equation

$$\rho \left(\frac{\partial V}{\partial t} + (V \cdot \nabla) V \right) = -\nabla P + \rho g$$

$$\rho \left(\frac{DV}{Dt} \right) = -\nabla P + \rho g$$

Bernoulli's Equation

Assumptions for Bernoulli's Equation

Before integrating, the following conditions must be met:

- 1. Steady Flow:** Fluid properties at any point do not change with time ($\frac{\partial V}{\partial t} = 0$).
- 2. Incompressible Flow:** The density of the fluid is constant ($\rho = C$).
- 3. Inviscid Flow:** The fluid is frictionless (zero viscosity).
- 4. Flow Along a Streamline:** The analysis is restricted to a single continuous streamline.

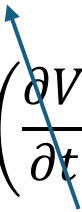
$$\rho \left(\frac{\partial V}{\partial t} + (V \cdot \nabla) V \right) = -\nabla P + \rho g \quad \Longrightarrow \quad \left(\frac{\partial V}{\partial t} + (V \cdot \nabla) V \right) = -\frac{\nabla P}{\rho} + g$$

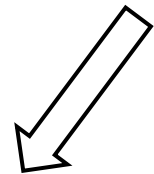
Apply Vector Identity to Advective Acceleration

$$(V \cdot \nabla) V = \nabla \left(\frac{V^2}{2} \right) - VX(\nabla \times V) = \nabla \left(\frac{V^2}{2} \right) - VX\omega$$

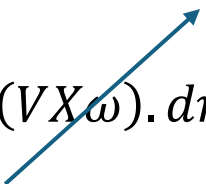
Bernoulli's Equation

Steady state


$$\left(\frac{\partial V}{\partial t} + \nabla \left(\frac{V^2}{2} \right) - VX\omega \right) = -\frac{\nabla P}{\rho} + g \quad \Longrightarrow \quad \left(0 + \nabla \left(\frac{V^2}{2} \right) - VX\omega \right) = -\nabla \left(\frac{P}{\rho} \right) - \nabla(gz)$$


$$\nabla \left(\frac{V^2}{2} + \frac{P}{\rho} + gz \right) = VX\omega$$

Integrate Along a Streamline


$$\nabla \left(\frac{V^2}{2} + \frac{P}{\rho} + gz \right) \cdot dr = (VX\omega) \cdot dr \quad \Longrightarrow \quad d \left(\frac{V^2}{2} + \frac{P}{\rho} + gz \right) = 0$$

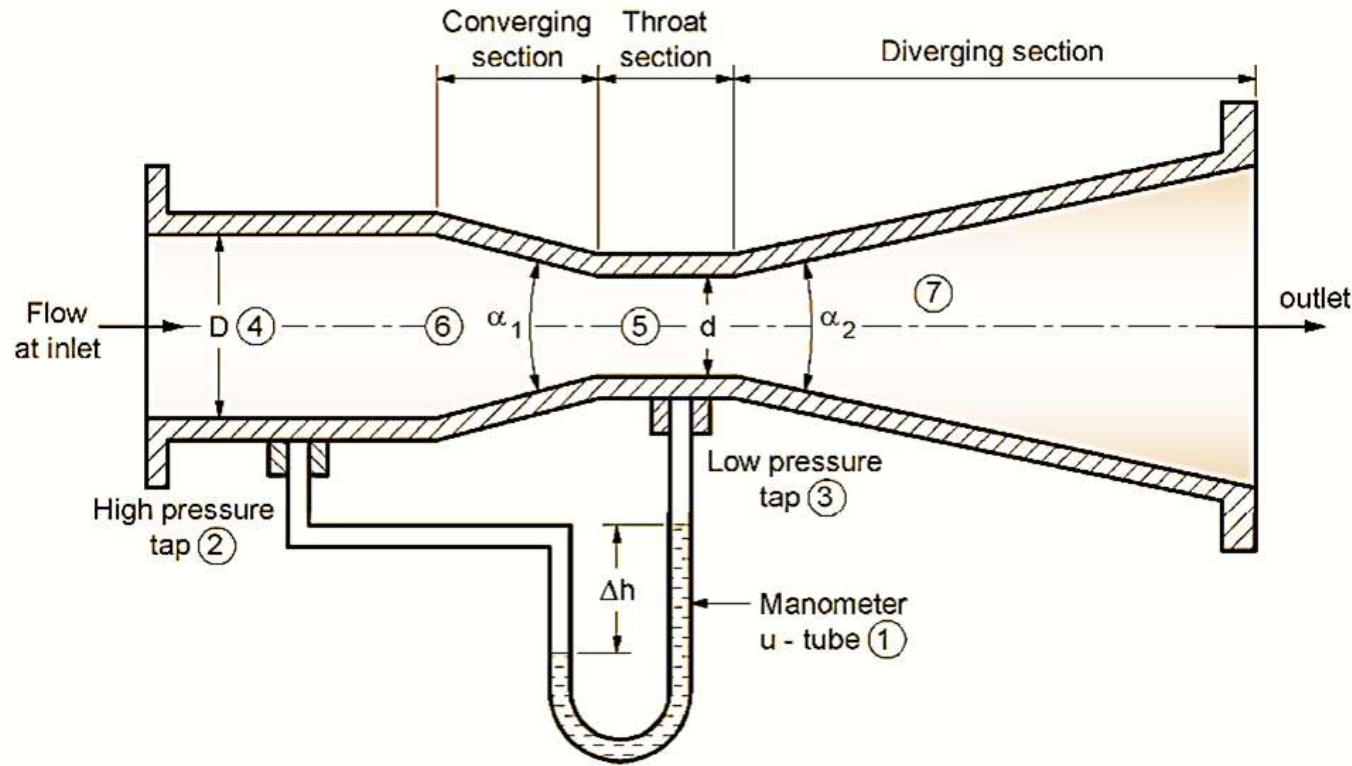
$$\frac{P}{\rho} + \frac{V^2}{2} + gz = C$$

Flow measurement devices

These devices calculate the total volumetric flow rate (Q) inside a pipe by deliberately restricting the flow area to create a measurable pressure drop, applying Bernoulli's principle.



Venturi meter flow rate derivation



$$\rho AV = C$$

$$\frac{P}{\rho} + \frac{V^2}{2} + gz = C$$

$$P = -\rho gz$$

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

$$\text{If } z_1 = z_2$$

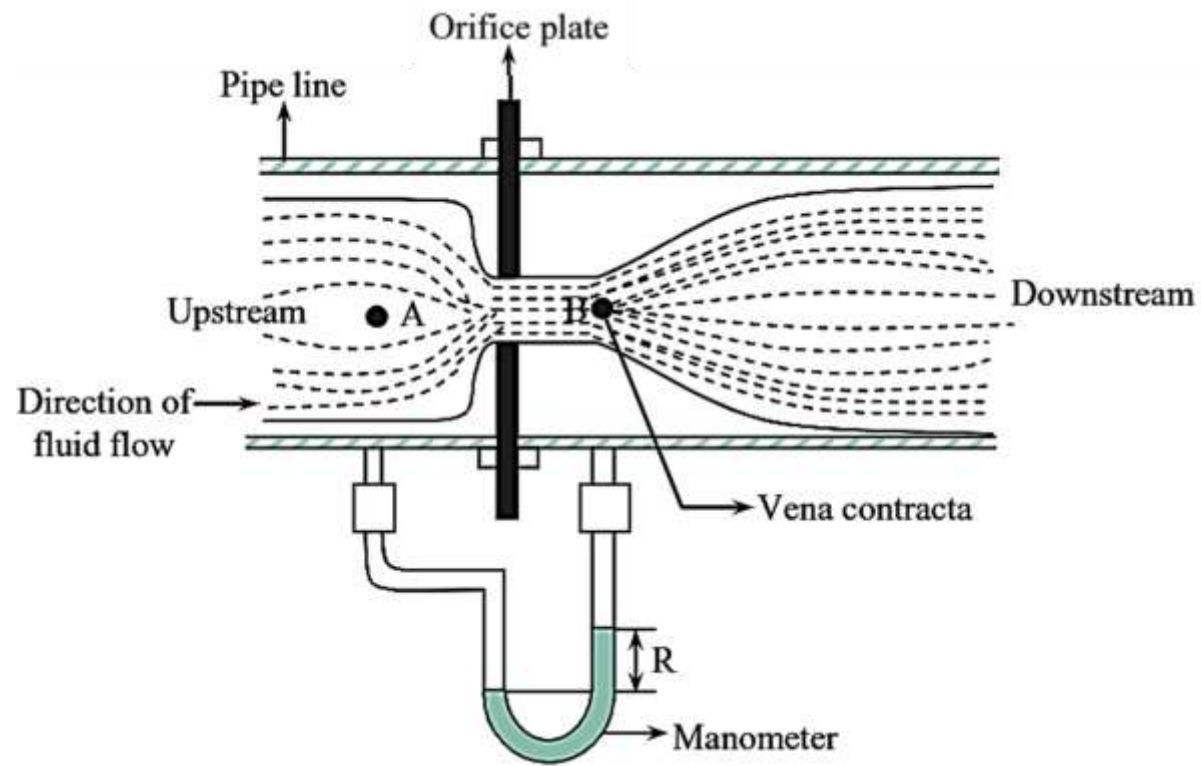
$$\frac{P_1 - P_2}{\rho g} = \frac{V_2^2 - V_1^2}{2g}$$

$$A_1 V_1 = A_2 V_2 \quad V_2 = \frac{A_1 V_1}{A_2}$$

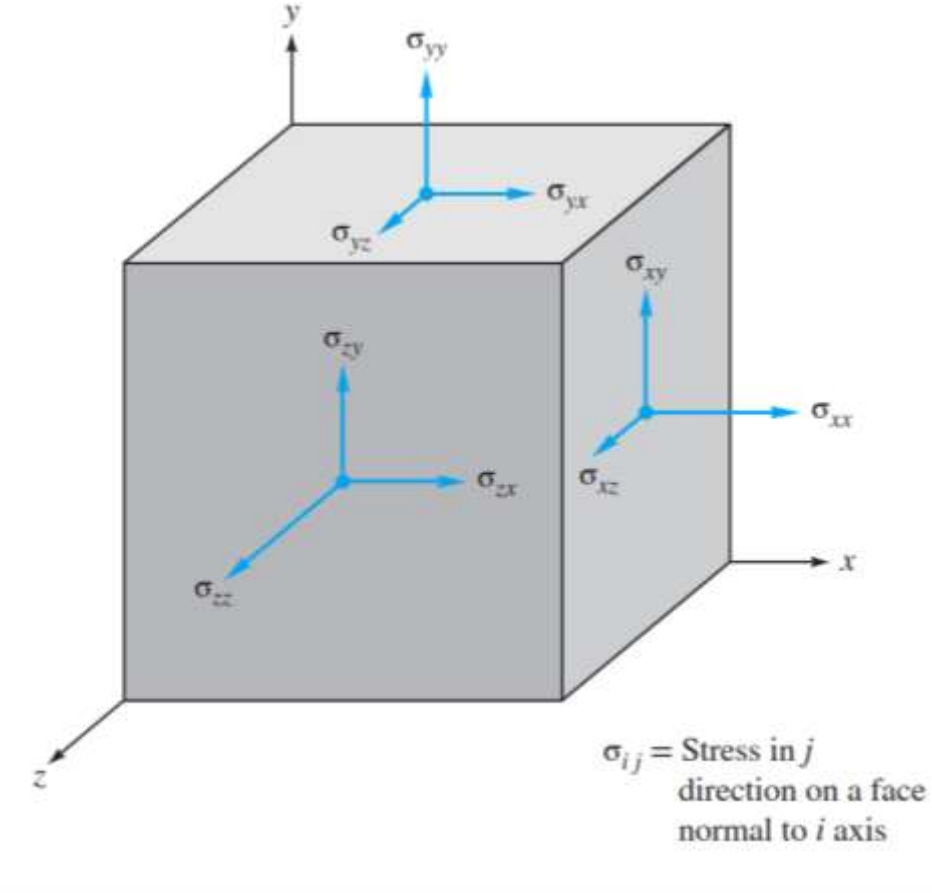
$$\frac{P_1 - P_2}{\rho g} = \frac{A_1^2 V_1^2 - A_2^2 V_1^2}{2g A_2^2} = \frac{(A_1^2 - A_2^2) V_1^2}{2g A_2^2}$$

$$Q_{th} = A_1 V_1$$

Orifice meter flow rate derivation



Navier Stokes Equation



$$\Sigma F = \frac{d}{dt} \left(\int_{CV} V \rho dV \right) + \left(\int_{CS} V \rho (V_r \cdot n) dA \right)$$

$$\Sigma F = \rho \left(\frac{\partial V}{\partial t} + (V \cdot \nabla) V \right)$$



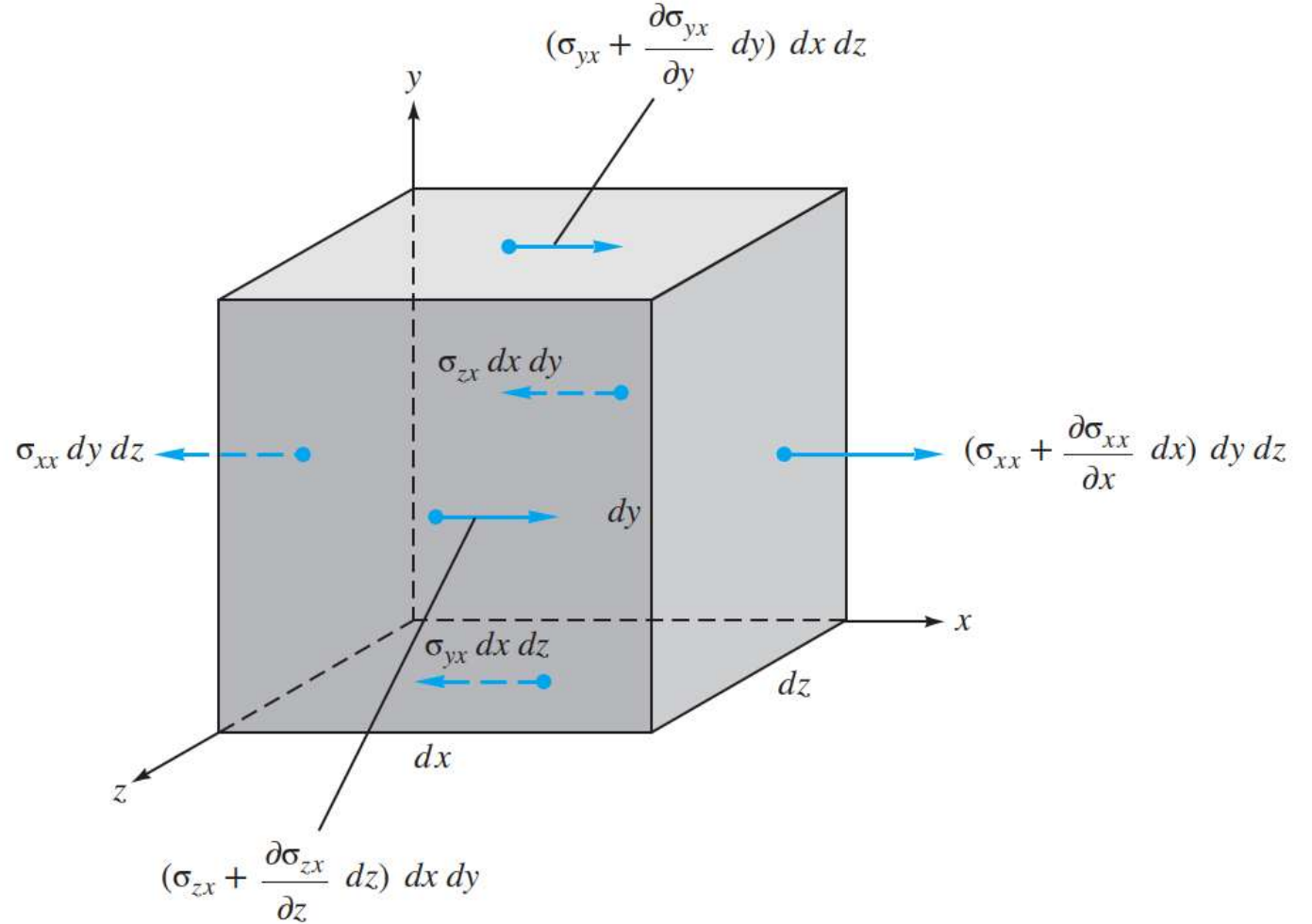
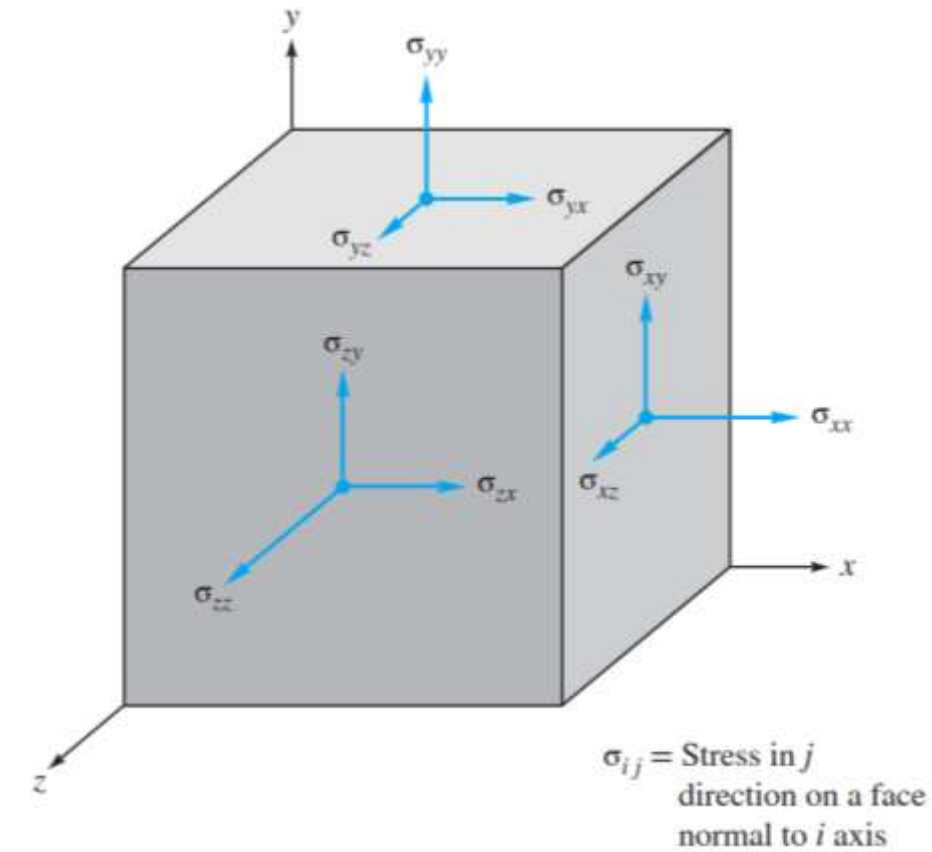
Body Forces + Surface forces

$$\Sigma F = F_B + F_S$$

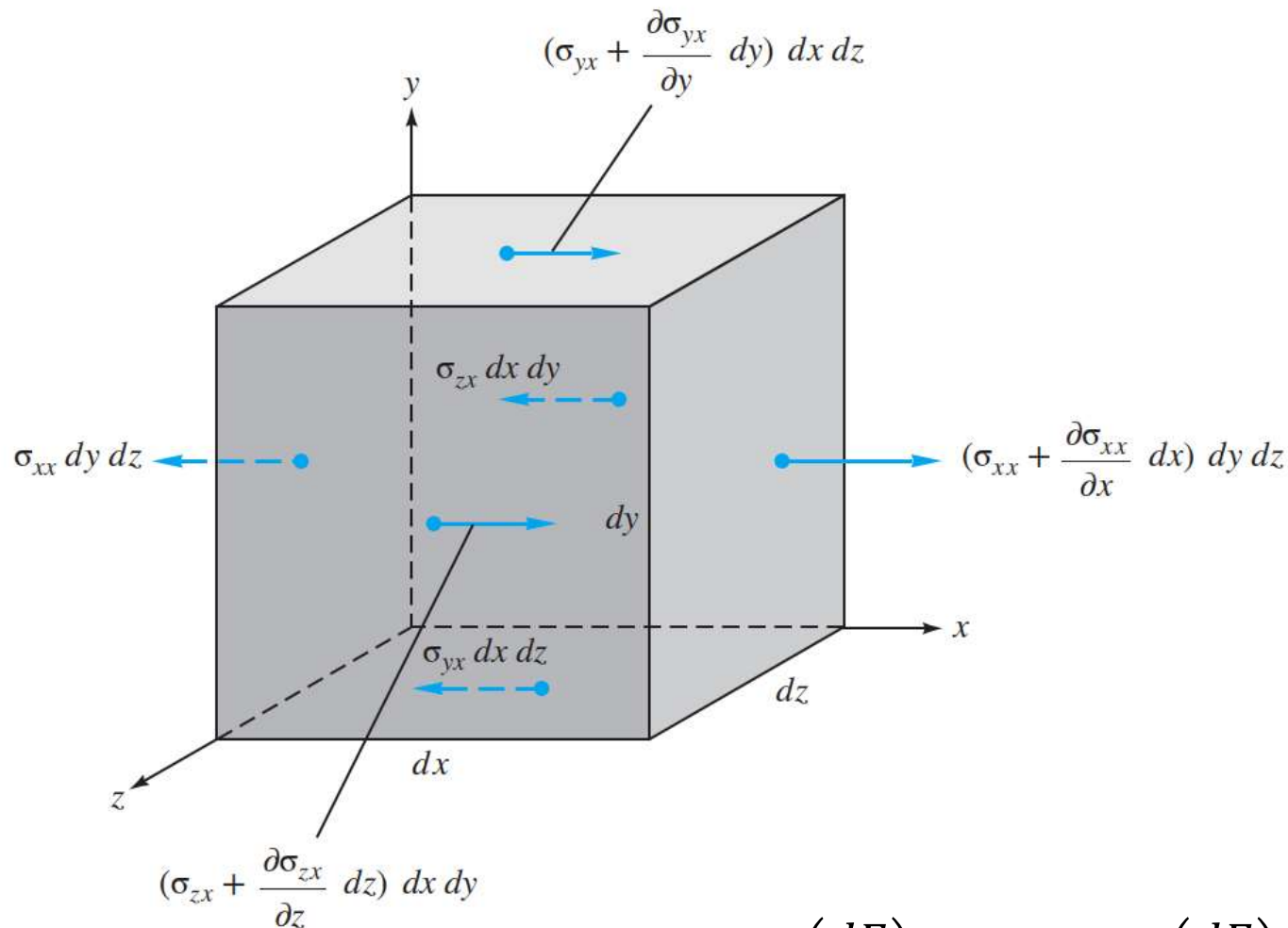
$$dF_B = dF_{gravity} = \rho g dx dy dz$$

$$dF_S = \text{forces due to stresses on surface}$$

Navier Stokes Equation



Navier Stokes Equation



$$dF_{x,s} = \left[\frac{\partial(\sigma_{xx})}{\partial x} + \frac{\partial(\sigma_{yx})}{\partial y} + \frac{\partial(\sigma_{zx})}{\partial z} \right] dx dy dz$$

$$\frac{dF_{x,s}}{dV} = -\frac{\partial p}{\partial x} + \frac{\partial(\sigma_{xx})}{\partial x} + \frac{\partial(\sigma_{yx})}{\partial y} + \frac{\partial(\sigma_{zx})}{\partial z}$$

$$\frac{dF_{y,s}}{dV} = -\frac{\partial p}{\partial y} + \frac{\partial(\sigma_{xy})}{\partial x} + \frac{\partial(\sigma_{yy})}{\partial y} + \frac{\partial(\sigma_{zy})}{\partial z}$$

$$\frac{dF_{z,s}}{dV} = -\frac{\partial p}{\partial z} + \frac{\partial(\sigma_{xz})}{\partial x} + \frac{\partial(\sigma_{yz})}{\partial y} + \frac{\partial(\sigma_{zz})}{\partial z}$$

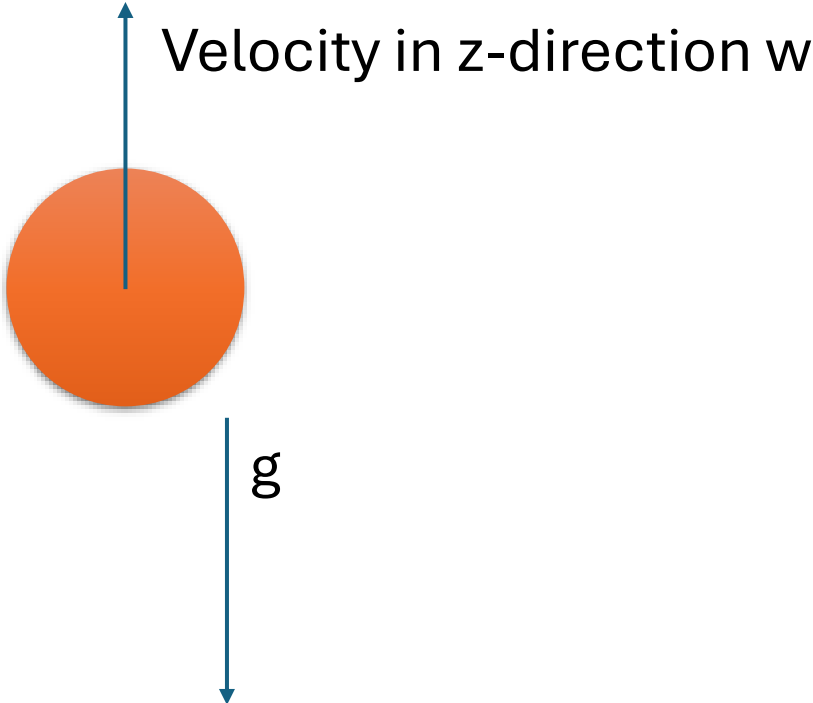
$$\left(\frac{dF}{dV} \right)_s = -\nabla p + \left(\frac{dF}{dV} \right)_{viscous} \quad \left(\frac{dF}{dV} \right)_{viscous} = \nabla \cdot \sigma_{ij}$$

Navier Stokes Equation

$$\rho \left(\frac{\partial V}{\partial t} + (V \cdot \nabla) V \right) = -\nabla p + \left(\frac{dF}{dV} \right)_{viscous} + \rho g$$

Dimensional Analysis

Basically, dimensional analysis is a method for reducing the number and complexity of experimental variables affecting a given physical phenomenon by using a compacting technique.

$$\frac{d^2 z}{dt^2} = -g$$
$$s = ut + \frac{1}{2}at^2$$
$$z = z_0 + w_0t - \frac{1}{2}gt^2$$


The diagram shows a solid orange circle representing an object. A blue arrow points vertically upwards from the center of the circle, labeled 'Velocity in z-direction w'. Another blue arrow points vertically downwards from the center of the circle, labeled 'g', representing the acceleration due to gravity.

Primary Dimension

- mass,
- length,
- time,
- temperature,
- electric current,
- amount of light,
- and amount of matter.

we have three available dimensional constants g , z_0 , and w_0 . We choose z_0 , and w_0 .
You can use g and z_0 , and/or with g and w_0

Dimensional Analysis

$$\frac{d^2 z}{dt^2} = -g \quad z^* = \frac{z}{z_0}; t^* = \frac{z_0 t}{w_0}$$

$$\frac{d^2(z^* z_0)}{d\left(\frac{w_0 t^*}{z_0}\right)^2} = -g$$

$$\frac{w_0^2 d^2(z^*)}{g z_0 d(t^*)^2} = -1 \quad Fr = \sqrt{\frac{w_0^2}{g z_0}}$$

$$\frac{d^2(z^*)}{d(t^*)^2} = -\frac{1}{Fr^2}$$

$$z^* = 1 + t^* - \frac{1}{2Fr^2} t^{*2}$$

Dimensional Analysis

$$z = z_0 + w_0 t - \frac{1}{2} g t^2$$

$$z^* = 1 + t^* - \frac{1}{2Fr^2} t^{*2} \quad Fr = \sqrt{\frac{w_0^2}{gz_0}}$$

Better insight into the relationships between key parameters.

for example, doubling w_0 has the same effect as decreasing z_0 by a factor of 4.

Second, it reduces the number of parameters in the problem.

As compared to the original one, the dependent variable, z ; one independent variable, t ; and three additional dimensional constants, g , w_0 , and z_0 .

The nondimensionalized problem contains one dependent parameter, z^* ; one independent parameter, t^* ; and only one additional parameter, namely the dimensionless Froude number, Fr .